Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 11
Chapter 1: Real Numbers

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. In a factor tree, if a node with value x branches into 3 and a node y, and node y branches into 5 and 7, then the value of x is: [1]
(a) 35 (b) 70 (c) 105 (d) 140

Q2. If the HCF of 408 and 1032 is expressible in the form $1032m - 408 \times 5$, then the value of m is: [1]
(a) 1 (b) 2 (c) 3 (d) 4

Q3. If two numbers are in the ratio 15 : 11 and their HCF is 13, then the numbers are: [1]
(a) 195 and 143 (b) 180 and 132 (c) 210 and 154 (d) 225 and 165

Q4. If $\text{HCF}(a, b) = 1$, then the numbers a and b are called: [1]
(a) twin primes (b) co-prime numbers (c) composite numbers (d) even numbers

Q5. The ratio between the LCM and HCF of 5, 15, and 20 is: [1]
(a) 9 : 1 (b) 4 : 3 (c) 11 : 1 (d) 12 : 1

Q6. **Assertion (A):** The number 4^n can never end with the digit 0 for any natural number $n \in \mathbb{N}$.
Reason (R): Any natural number ending with the digit 0 must have both 2 and 5 in its prime factorisation. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. In a factor tree, a number N decomposes into 2 and A; A decomposes into 3 and B; B decomposes into 5 and 13. Find the values of N, A, and B. [2]

Q8. Find the smallest natural number by which $\sqrt{27}$ should be multiplied so as to get a rational number. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Find the least positive integer which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4, and 3. [3]

Q10. Two tankers contain 850 litres and 680 litres of petrol respectively. Find the maximum capacity of a container which can measure the petrol of either tanker in an exact number of times. [3]

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

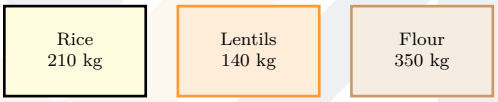
Q11. Let n be any positive integer. (i) Prove that $n(n+1)(n+2)$ is always divisible by 6. (ii) Prove that $n^5 - n$ is divisible by 5. (iii) Hence, prove that $n^5 - n$ is always divisible by 30 for all $n \in \mathbb{N}$. [5]

Q12. A student constructs an extended factor tree for a composite number M. The tree decomposes as follows: $M = 2 \times N_1$; $N_1 = 3 \times N_2$; $N_2 = 3 \times N_3$; $N_3 = 5 \times N_4$; $N_4 = 7 \times 11$. (i) Find the values of N_4, N_3, N_2, N_1, and M. (ii) Express M in canonical prime factorisation form. (iii) Determine the total number of positive factors of M and calculate the sum of all exponents in its prime factorisation. [5]

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: CASE STUDY 11: HUMANITARIAN DISASTER RELIEF RATION KIT PACKAGING [4 Marks]

A disaster management agency is preparing emergency relief family food packets following floods in Assam. The central depot has received donations of 210 kg of rice, 140 kg of lentils (dal), and 350 kg of wheat flour. Relief workers must pack these provisions into identical emergency relief kits such that each kit contains an exact equal integer amount of each staple, and no food grains are left unpackaged.



Max Identical Kits = **HCF**$(210, 140, 350) = 70$

(i) Find the maximum number of identical relief kits that can be prepared. [2]

(ii) How many kilograms of rice, lentils, and flour will each family kit contain? [1]

(iii) Verify that $\text{HCF}(210, 140) \times \text{LCM}(210, 140) = 210 \times 140$. [1]

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